

# Inhibition-gated sparsity in trained spiking networks

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## 1 Abstract

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## 2 Introduction

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## 3 Methods

### 3.1 The neuron model

This paper uses a conductance-based leaky integrate-and-fire (LIF) network with separate excitatory and inhibitory populations, fixed recurrent E $\leftrightarrow$ I matrices, and exponential synapses. The COBA neurons obey

$$C_m \frac{dV}{dt} = -g_L(V - E_L) - g_e(V - E_e) - g_i(V - E_i) \quad (1)$$

where  $V$  is membrane potential,  $t$  is time,  $C_m$  is capacitance,  $g_L$  is leak conductance,  $g_e$  is the excitatory synaptic conductance and  $g_i$  is the inhibitory synaptic conductance with reversal potentials  $E_L = -65$  mV (leak),  $E_e = 0$  mV (excitatory),  $E_i = -80$  mV (inhibitory). Each conductance term pulls  $V$  toward its reversal potential at a rate proportional to the conductance. Excitatory synapses drive  $V$  up towards  $E_e = 0$  and inhibitory synapses both pull it down towards  $E_i = -80$  and increase total conductance, which lowers the effective membrane time constant - this is *shunting inhibition* and is required to create the PING oscillations.

Equation 1 can be written as a linear first-order ODE in the form  $C_m \dot{V} = -AV + B$  with  $A = g_L + g_e + g_i$  and  $B = g_LE_L + g_eE_e + g_iE_i$ . Setting  $\dot{V} = 0$  we find the voltage in which the inflow  $B$  exactly balances the outflow  $AV$ , and name that voltage  $V_\infty$ . We define accordingly:

$$g_{\text{tot}} = g_L + g_e + g_i \quad (2)$$

$$\tau_{\text{eff}} = \frac{C_m}{g_{\text{tot}}} \quad (3)$$

$$V_\infty = \frac{g_LE_L + g_eE_e + g_iE_i}{g_{\text{tot}}} \quad (4)$$

Substituting Equations (2)–(4) into Equation 1 and dividing through by  $g_{\text{tot}}$  yields the standard “decay-to-steady-state” form

$$\tau_{\text{eff}} \frac{dV}{dt} = -(V - V_\infty) \quad (5)$$

Integrating over an interval where  $\tau_{\text{eff}}$  and  $V_\infty$  are held constant gives closed form solution

$$V(t) = V_\infty + (V_0 - V_\infty) \exp\left(\frac{-t}{\tau_{\text{eff}}}\right), \quad (6)$$

the membrane interpolates from its starting value  $V_0$  toward  $V_\infty$  on timescale  $\tau_{\text{eff}}$ . However in continuous time  $\tau_{\text{eff}}$  and  $V_\infty$  are *not* constant, both depend on  $g_{e(t)}$  and  $g_{i(t)}$  which evolve as spikes arrive. A zero-order hold makes this numerically tractable, freeze  $g_e$  and  $g_i$  over one timestep  $\Delta t$ , integrate Equation 5 exactly under that freeze, then update  $g_e$  and  $g_i$  for the next timestep, then Equation 5 integrates analytically to

$$V_{t+1} = V_\infty + (V_t - V_\infty) \exp\left(\frac{-\Delta t}{\tau_{\text{eff}}}\right). \quad (7)$$

After the membrane update, apply the spike reset:

$$S_{t+1} = \mathbf{1}[V_{t+1} \geq V_{\text{th}}], V_{t+1} \leftarrow V_{\text{reset}} \text{ if } S_{t+1} = 1 \text{ or refractory} \quad (8)$$

with  $V_{\text{th}} = -50$  mV and  $V_{\text{reset}} = -65$  mV. Refractory periods are  $\tau_{\text{ref}}^e = 3$  ms and  $\tau_{\text{ref}}^i = 1.5$  ms, held in a per-neuron countdown that suppresses spiking until exhaustion.

Each conductance is the convolution of its presynaptic spike train with a decaying exponential. For a single neuron receiving feedforward input only:

$$g_{e,t+1} = \alpha g_{e,t} + W_{\text{in}} s_t^{\text{inp}} \quad (9)$$

where  $\alpha = \exp\left(\frac{-\Delta t}{\tau_{\text{AMPA}}}\right)$  is the per-step AMPA decay factor. Between spikes the conductance decays exponentially toward zero; each arriving spike adds a kick proportional to the synaptic weight. The inhibitory conductance  $g_i$  follows the same form with GABA decay  $\gamma = \exp\left(\frac{-\Delta t}{\tau_{\text{GABA}}}\right)$ .

In a network with E and I populations, two recurrent pathways close the PING loop alongside the feedforward input. There are no E $\rightarrow$ E connections.

$$g_{e,t+1}^E = \alpha g_{e,t}^E + W_{\text{in}} s_t^{\text{inp}} \quad (10)$$

$$g_{i,t+1}^E = \gamma (g_{i,t}^E + W_{\text{ie}} s_t^i) \quad (11)$$

$$g_{e,t+1}^E = \gamma (g_{e,t}^E + W_{\text{ei}} s_t^e) \quad (12)$$

Equation 10 is the excitatory conductance on E neurons — purely feedforward, no recurrent E $\rightarrow$ E term. Equation 11 is the inhibitory conductance on E neurons — driven by I spikes through  $W_{\text{ie}}$ . Equation 12 is the excitatory conductance on I neurons — driven by

E spikes through  $W_{ei}$ . I neurons receive no inhibition. The trainable feedforward input  $W_{in}s_t^{\text{inp}}$  enters E's excitatory conductance after the decay step, arriving fresh each timestep regardless of recurrent state.

The model emits two modes, COBA and PING, when PING mode is active the  $W_{ei}$  and  $W_{ie}$  are fixed at init and are *not* trainable.  $W_{ei}$  and  $W_{ie}$  are controlled by a single scalar flag  $s$ :

$$W_{ei} \sim N(s, 0.1s) \quad (13)$$

$$W_{ie} \sim N(2.0s, 0.2s) \quad (14)$$

At  $s = 0$  both matrices are zero, the loop is open, and the network is feedforward COBA. At  $s = 1$  the default initialization gives  $W_{ei} \approx 1\mu\text{S}$  and  $W_{ie} \approx 2\mu\text{S}$ , strong enough to sustain the PING rhythm from the first forward pass of training.

## 3.2 The network model

## 3.3 Training

## 4 Results

### 4.1 Lower rates

### 4.2 Control

### 4.3 Perturbation

## 5 Discussion

### 5.1 Why PING has a rate floor

## 6 Conclusion

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